Q1. JEE Main 2026 (21 January Shift 2)

Let the line L pass through the point $(-3,5,2)$ and make equal angles with the positive coordinate axes. If the distance of L from the point $(-2, \mathrm{r}, 1)$ is $\sqrt{\frac{14}{3}}$, then the sum of all possible values of r is :

(1) 10 (2) 6 (3) 12 (4) 16

Q2. JEE Main 2026 (21 January Shift 2)

Let the line L_1 be parallel to the vector $-3\hat{i}+2\hat{j}+4\hat{k}$ and pass through the point (2,6,7), and the line L_2 be parallel to the vector $2\hat{i}+\hat{j}+3\hat{k}$ and pass through the point $(4,3,5)$. If the line L_3 is parallel to the vector $-3\hat{i}+5\hat{j}+16\hat{k}$ and intersects the lines L_1 and L_2 at the points C and D, respectively, then $|\overrightarrow{CD}|^2$ is equal to :

(1) 89 (2) 312 (3) 171 (4) 290

Q3. JEE Main 2026 (23 January Shift 1)

Let the direction cosines of two lines satisfy the equations : $4l+m-n=0$ and $2mn+10nl+3lm=0$. Then the cosine of the acute angle between these lines is :

(1) $\frac{20}{3\sqrt{38}}$ (2) $\frac{10}{3\sqrt{38}}$ (3) $\frac{10}{7\sqrt{38}}$ (4) $\frac{10}{\sqrt{38}}$

Q4. JEE Main 2026 (23 January Shift 1)

The vertices B and C of a triangle ABC lie on the line $\frac{x}{1}=\frac{1-y}{-2}=\frac{z-2}{3}$. The coordinates of A and B are $(1,6,3)$ and $(4,9,\alpha)$ respectively and C is at a distance of 10 units from B. The area (in sq. units) of $\triangle$ABC is :

(1) $20\sqrt{13}$ (2) $5\sqrt{13}$ (3) $15\sqrt{13}$ (4) $10\sqrt{13}$

Q5. JEE Main 2026 (28 January Shift 2)

Let $\mathrm{Q}(\mathrm{a}, \mathrm{b}, \mathrm{c})$ be the image of the point $\mathrm{P}(3,2,1)$ in the line $\frac{x-1}{1}=\frac{y}{2}=\frac{z-1}{1}$. Then the distance of Q from the line $\frac{x-9}{3}=\frac{y-9}{2}=\frac{z-5}{-2}$ is

(1) 8 (2) 5 (3) 7 (4) 6

Q6. JEE Main 2026 (28 January Shift 2)

If the distance of the point $\mathrm{P}(43,\alpha,\beta), \beta<0$, from the line $\vec{\mathrm{r}}=4\hat{i}-\hat{k}+\mu(2\hat{i}+3\hat{k}), \mu \in \mathbf{R}$ along a line with direction ratios $3,-1,0$ is $13\sqrt{10}$, then $\alpha^2+\beta^2$ is equal to ____

Q7. JEE Main 2026 (24 January Shift 2)

The sum of all values of α, for which the shortest distance between the lines $\frac{x+1}{\alpha}=\frac{y-2}{-1}=\frac{z-4}{-\alpha}$ and $\frac{x}{\alpha}=\frac{y-1}{2}=\frac{z-1}{2\alpha}$ is $\sqrt{2}$, is

(1) 6 (2) 8 (3) -8 (4) -6

Q8. JEE Main 2026 (22 January Shift 2)

Let L be the line $\frac{x+1}{2}=\frac{y+1}{3}=\frac{z+3}{6}$ and let S be the set of all points $(\mathrm{a}, \mathrm{b}, \mathrm{c})$ on L, whose distance from the line $\frac{x+1}{2}=\frac{y+1}{3}=\frac{z-9}{0}$ along the line L is 7. Then $\sum_{(a,b,c)\in S}(a+b+c)$ is equal to :

(1) 6 (2) 34 (3) 40 (4) 28

Q9. JEE Main 2026 (22 January Shift 1)

If the image of the point $\mathrm{P}(1,2,a)$ in the line $\frac{x-6}{3}=\frac{y-7}{2}=\frac{7-z}{2}$ is $\mathrm{Q}(5,b,\mathrm{c})$, then $a^2+b^2+c^2$ is equal to

(1) 298 (2) 293 (3) 264 (4) 283

Q10. JEE Main 2026 (22 January Shift 1)

Let $\mathrm{P}(\alpha,\beta,\gamma)$ be the point on the line $\frac{x-1}{2}=\frac{y+1}{-3}=z$ at a distance $4\sqrt{14}$ from the point $(1,-1,0)$ and nearer to the origin. Then the shortest distance, between the lines $\frac{x-\alpha}{1}=\frac{y-\beta}{2}=\frac{z-\gamma}{3}$ and $\frac{x+5}{2}=\frac{y-10}{1}=\frac{z-3}{1}$, is equal to

(1) $4\sqrt{\frac{7}{5}}$ (2) $2\sqrt{\frac{7}{4}}$ (3) $7\sqrt{\frac{5}{4}}$ (4) $4\sqrt{\frac{5}{7}}$

Q11. JEE Main 2026 (23 January Shift 2)

If the image of the point $\mathrm{P}(a,2,a)$ in the line $\frac{x}{2}=\frac{y+a}{1}=\frac{z}{1}$ is Q and the image of Q in the line $\frac{x-2b}{2}=\frac{y-a}{1}=\frac{z+2b}{-5}$ is P, then $a+b$ is equal to _____.

Q12. JEE Main 2026 (21 January Shift 1)

Let (α,β,γ) be the co-ordinates of the foot of the perpendicular drawn from the point $(5,4,2)$ on the line $\vec{r}=(-\hat{i}+3\hat{j}+\hat{k})+\lambda(2\hat{i}+3\hat{j}-\hat{k})$.

Then the length of the projection of the vector $\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k}$ on the vector $6\hat{i}+2\hat{j}+3\hat{k}$ is :

(1) $\frac{15}{7}$ (2) 4 (3) $\frac{18}{7}$ (4) 3

Q13. JEE Main 2026 (28 January Shift 1)

If the distances of the point $(1,2,a)$ from the line $\frac{x-1}{1}=\frac{y}{2}=\frac{z-1}{1}$ along the lines $\mathrm{L}_1:\frac{x-1}{3}=\frac{y-2}{4}=\frac{z-a}{b}$ and $\mathrm{L}_2:\frac{x-1}{1}=\frac{y-2}{4}=\frac{z-a}{c}$ are equal, then $a+b+c$ is equal to

(1) 5 (2) 7 (3) 4 (4) 6

Q14. JEE Main 2026 (24 January Shift 1)

Let the lines $\mathrm{L}_1:\vec{r}=\hat{\mathrm{i}}+2\hat{\mathrm{j}}+3\hat{\mathrm{k}}+\lambda(2\hat{\mathrm{i}}+3\hat{\mathrm{j}}+4\hat{\mathrm{k}}),\lambda\in\mathbb{R}$ and $\mathrm{L}_2:\vec{r}=(4\hat{\mathrm{i}}+\hat{\mathrm{j}})+\mu(5\hat{\mathrm{i}}+2\hat{\mathrm{j}}+\hat{\mathrm{k}}),\mu\in\mathbb{R}$, intersect at the point R. Let P and Q be the points lying on lines L_1 and L_2, respectively, such that $|\overrightarrow{\mathrm{PR}}|=\sqrt{29}$ and $|\overrightarrow{\mathrm{PQ}}|=\sqrt{\frac{47}{3}}$. If the point P lies in the first octant, then $27(\mathrm{QR})^2$ is equal to

(1) 348 (2) 340 (3) 320 (4) 360

Q15. JEE Main 2026 (24 January Shift 1)

Let a line L passing through the point $\mathrm{P}(1,1,1)$ be perpendicular to the lines $\frac{x-4}{4}=\frac{y-1}{1}=\frac{z-1}{1}$ and $\frac{x-17}{1}=\frac{y-71}{1}=\frac{z}{0}$. Let the line L intersect the $yz-$ plane at the point Q. Another line parallel to L and passing through the point $\mathrm{S}(1,0,-1)$ intersects the yz-plane at the point R. Then the square of the area of the parallelogram PQRS is equal to ____.

ANSWERS AND SOLUTIONS

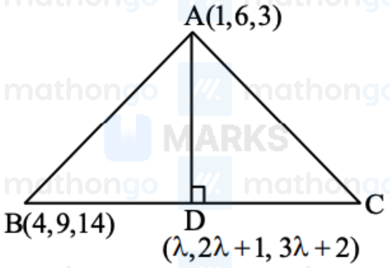
1. (1)	**2.** (4)	**3.** (2)	**4.** (2)	**5.** (3)	**6.** 170	**7.** (4)	**8.** (2)
9. (1)	**10.** (1)	**11.** 3	**12.** (3)	**13.** (2)	**14.** (4)	**15.** 6	

1. **(1)** Line L through $(-3,5,2)$ making equal angles with axes has direction $(1,1,1)$.
Point $Q=(-2,r,1)$, $\vec{PQ}=(1,r-5,-1)$.
$\vec{PQ}\times\vec{d}=(r-4,-2,6-r)$.
$|\vec{PQ}\times\vec{d}|^2=(r-4)^2+4+(6-r)^2=2r^2-20r+56$.
Distance $=\sqrt{\frac{2r^2-20r+56}{3}}=\sqrt{\frac{14}{3}}$.
$2r^2-20r+56=14\Rightarrow r^2-10r+21=0\Rightarrow r=3$ or $r=7$.
Sum $=3+7=10$.

2. **(4)** $L_1:\dfrac{x-2}{-3}=\dfrac{y-6}{2}=\dfrac{z-7}{4}$
Point C on L_1: $(-3\lambda_1+2,2\lambda_1+6,4\lambda_1+7)$
$L_2:\dfrac{x-4}{2}=\dfrac{y-3}{1}=\dfrac{z-5}{3}$
Point D on L_2: $(2\lambda_2+4,\lambda_2+3,3\lambda_2+5)$
Dr's of line L_3 :
$L_3:\dfrac{2\lambda_2+3\lambda_1+2}{-3}=\dfrac{\lambda_2-2\lambda_1-3}{5}=\dfrac{3\lambda_2-4\lambda_1-2}{16}$
$\lambda_1=-3,\lambda_2=2$
$C(11,0,-5)$
$D(8,5,11)$
$|\overrightarrow{CD}|^2=3^2+5^2+16^2=290$

3. **(2)** From $n=4l+m$, substitute in $2mn+10nl+3lm=0$:
$40l^2+21lm+2m^2=0$. Let $t=l/m$: $40t^2+21t+2=0$.
$t=\frac{-21\pm11}{80}$, giving $t=-1/8$ or $t=-2/5$.
$l/m=-1/8$: direction $(l,m,n)=(-1,8,4)$.
$l/m=-2/5$: direction $(l,m,n)=(-2,5,-3)$.
$\cos\theta=\frac{|2+40-12|}{\sqrt{81}\sqrt{38}}=\frac{30}{9\sqrt{38}}=\frac{10}{3\sqrt{38}}$.

4. **(2)** $\frac{4}{1} = \frac{9-1}{2} = \frac{\alpha - 2}{3} \Rightarrow \alpha = 14$



$\overrightarrow{AD} \cdot (\hat{i} + 2\hat{j} + 3\hat{k}) = 0$

$(\lambda - 1)\hat{i} + (2\lambda - 5)\hat{j} + (3\lambda - 1)\hat{k} = \overrightarrow{AD}$

$\Rightarrow \lambda - 1 + 4\lambda - 10 + 9\lambda - 3 = 0$

$\Rightarrow 14\lambda = 14 \Rightarrow \lambda = 1$

$D = (1, 3, 5)$

$AD = \sqrt{3^2 + 2^2} = \sqrt{13}$

$Ar(\triangle ABC) = \frac{1}{2} \times \sqrt{13} \times 10 = 5\sqrt{13}$

5. **(3)** Line 1 passes through $(1, 0, 1)$ with direction $(1, 2, 1)$. Point on line: $(1 + t, 2t, 1 + t)$.

For projection Q of P(3, 2, 1): $(t - 2, 2t - 2, t) \cdot (1, 2, 1) = 0$ gives $6t = 6$, so $t = 1$.

Thus Q = (2, 2, 2).

Distance from Q to line 2 (passing through $(9, 9, 5)$ with direction $(3, 2, -2)$):

$\vec{QR} = (7, 7, 3)$.

$\vec{QR} \times \vec{d} = -20\vec{i} + 23\vec{j} - 7\vec{k}$.

$|\vec{QR} \times \vec{d}| = \sqrt{400 + 529 + 49} = \sqrt{978}$.

$|\vec{d}| = \sqrt{17}$.

Distance = $\frac{\sqrt{978}}{\sqrt{17}} = \sqrt{57.53} \approx 7.59 \approx 7$.

6. **(170)** Point P(43, α, β) has distance $13\sqrt{10}$ from line $\vec{r} = 4\vec{i} - \vec{k} + \mu(2\vec{i} + 3\vec{k})$.

The line passes through A(4, 0, -1) with direction $\vec{d} = (2, 0, 3)$.

Using distance formula: $d = \frac{|\vec{AP} \times \vec{d}|}{|\vec{d}|}$.

With $\vec{AP} = (39, \alpha, \beta + 1)$, the cross product gives $\vec{AP} \times \vec{d} = (3\alpha, 2\beta - 115, -2\alpha)$.

Thus $|\vec{AP} \times \vec{d}| = \sqrt{13\alpha^2 + (2\beta - 115)^2}$ and $|\vec{d}| = \sqrt{13}$.

Setting $\frac{\sqrt{13\alpha^2 + (2\beta - 115)^2}}{\sqrt{13}} = 13\sqrt{10}$ gives $\alpha^2 + \frac{(2\beta - 115)^2}{13} = 1690$.

Solving with the constraint β < 0: α = 5, β = -13, hence $\alpha^2 + \beta^2 = 25 + 169 = 170$.

7. **(4)** The given lines are $L_1 : \frac{x+1}{\alpha} = \frac{y-2}{-1} = \frac{z-4}{-\alpha}$ and $L_2 : \frac{x}{\alpha} = \frac{y-1}{2} = \frac{z-1}{2\alpha}$.

For L_1, point $\vec{a}_1 = (-1, 2, 4)$ and direction $\vec{b}_1 = (\alpha, -1, -\alpha)$.

For L_2, point $\vec{a}_2 = (0, 1, 1)$ and direction $\vec{b}_2 = (\alpha, 2, 2\alpha)$.

The shortest distance d is given by $d = \frac{|(\vec{a}_2 - \vec{a}_1)\cdot(\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$.

$\vec{a}_2 - \vec{a}_1 = (1, -1, -3)$.

$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \alpha & -1 & -\alpha \\ \alpha & 2 & 2\alpha \end{vmatrix} = \hat{i}(-2\alpha + 2\alpha) - \hat{j}(2\alpha^2 + \alpha^2) + \hat{k}(2\alpha + \alpha) = (0, -3\alpha^2, 3\alpha)$.

$|\vec{b}_1 \times \vec{b}_2| = \sqrt{0 + 9\alpha^4 + 9\alpha^2} = 3|\alpha|\sqrt{\alpha^2 + 1}$.

$(\vec{a}_2 - \vec{a}_1)\cdot(\vec{b}_1 \times \vec{b}_2) = (1)(0) + (-1)(-3\alpha^2) + (-3)(3\alpha) = 3\alpha^2 - 9\alpha$.

Given $d = \sqrt{2}$, so $\frac{|3\alpha^2 - 9\alpha|}{3|\alpha|\sqrt{\alpha^2+1}} = \sqrt{2} \Rightarrow \frac{|\alpha - 3|}{\sqrt{\alpha^2+1}} = \sqrt{2}$.

Squaring both sides: $(\alpha - 3)^2 = 2(\alpha^2 + 1) \Rightarrow \alpha^2 - 6\alpha + 9 = 2\alpha^2 + 2$.

$\alpha^2 + 6\alpha - 7 = 0$.

The values of α are the roots of this quadratic equation.

Sum of values of $\alpha = -\frac{6}{1} = -6$.

8. **(2)** M is the point of intersection of L_1 & L_2

$\Rightarrow 2\lambda - 1 = 2\mu - 1, 3\lambda - 1 = 3\mu - 1, 6\lambda - 3 = 9$

$\Rightarrow \lambda = 2 = \mu$

$\Rightarrow M(3, 5, 9)$

Now let point P be $(2K - 1, 3K - 1, 6K - 3)$ on L_2

such that $PM = 7$

$\Rightarrow \sqrt{(2K - 4)^2 + (3K - 6)^2 + (6K - 12)^2} = 7$

$\Rightarrow 49K^2 + 196 - 196K = 49$

$\Rightarrow K^2 + 4 - 4K = 1$

$\Rightarrow K^2 - 4K + 3 = 0$

$\Rightarrow K = 1, 3$

So points P & Q are $(1, 2, 3)$ & $(5, 8, 15)$

So sum of all co-ordinates of P & Q $= 34$

9. **(1)** Line: point $(6, 7, 7)$, direction $(3, 2, -2)$.

General point on line: $(6 + 3t, 7 + 2t, 7 - 2t)$.

Foot of perpendicular from $P(1, 2, a)$: $\vec{PM} \cdot (3, 2, -2) = 0$.

$3(5 + 3t) + 2(5 + 2t) - 2(7 - 2t - a) = 0$

$\Rightarrow 17t + 11 + 2a = 0$.

Image $Q = 2M - P = (11 + 6t, 12 + 4t, 14 - 4t - a)$.

Given $Q_x = 5$: $11 + 6t = 5 \Rightarrow t = -1$.

From $17(-1) + 11 + 2a = 0 \Rightarrow a = 3$.

$b = 12 + 4(-1) = 8$, $c = 14 - 4(-1) - 3 = 15$.

$a^2 + b^2 + c^2 = 9 + 64 + 225 = 298$.

10. (1) Let $P(2\lambda+1,-3\lambda-1,\lambda)$

Then $4\lambda^2+9\lambda^2+\lambda^2=16\cdot 14\Rightarrow\lambda=\pm 4\Rightarrow -4$

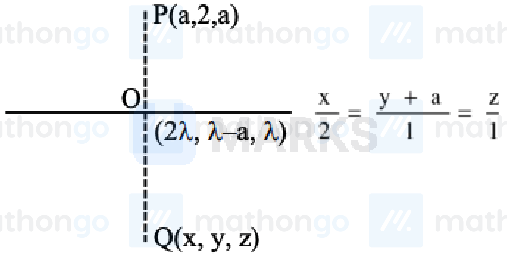
(nearer to origin)

$\therefore P(-7,11,-4)$

$$\therefore \text{Shortest distance}=\frac{\begin{vmatrix}2 & -1 & 7\\ 1 & 2 & 3\\ 2 & 1 & 1\end{vmatrix}}{\left|\begin{vmatrix}\hat{i} & \hat{j} & k\\ 1 & 2 & 3\\ 2 & 1 & 1\end{vmatrix}\right|}$$

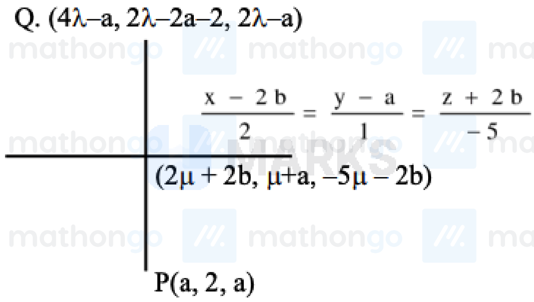
$$=\frac{28}{\sqrt{1+25+9}}=\frac{4\sqrt{7}}{\sqrt{5}}$$

11.



(3)

Now,



$$\frac{2\lambda-a+a}{2}=-5\mu-2b$$

and $\dfrac{a+4\lambda-a}{2}=2\mu+2b$

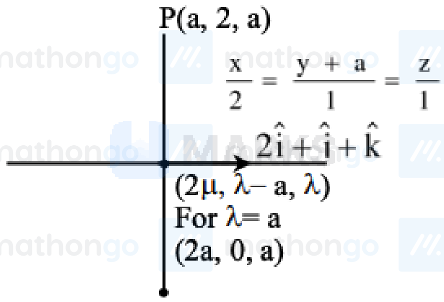
and $\dfrac{2\lambda-2a-2+2}{2}=\mu+a$

$\Rightarrow \lambda-\mu=b$

$\lambda-\mu=2a$

$\lambda+5\mu=-2b$

$\Rightarrow b=2a$ and $\lambda=a,\mu=-a$



$(a\hat{i} - 2\hat{j} + 0\hat{k}) \cdot (2\hat{i} + \hat{j} + \hat{k}) = 0$

$2a - 2 = 0$

$a = 1$

$b = 2a = 2 \times 1 = 2$

$a + b = 1 + 2 = 3$

12. (3) Line: $\vec{r} = (-1, 3, 1) + \lambda(2, 3, -1)$, Point P = (5, 4, 2).

Any point on line: $Q = (-1 + 2\lambda, 3 + 3\lambda, 1 - \lambda)$.

$\vec{PQ} = (-6 + 2\lambda, -1 + 3\lambda, -1 - \lambda)$.

For perpendicular: $\vec{PQ} \cdot (2, 3, -1) = 0$ gives $14\lambda - 14 = 0$, so $\lambda = 1$.

Foot $(\alpha, \beta, \gamma) = (1, 6, 0)$.

Projection of $(1, 6, 0)$ on $(6, 2, 3)$: $\frac{|6+12+0|}{\sqrt{49}} = \frac{18}{7}$.

13. (2) Line L: $\frac{x-1}{1} = \frac{y}{2} = \frac{z-1}{1}$

L_1: $\frac{x-1}{3} = \frac{y-2}{4} = \frac{z-a}{b} = \lambda$

L_2: $\frac{x-1}{1} = \frac{y-2}{4} = \frac{z-a}{c} = \mu$

Point $A(3\lambda + 1, 4\lambda + 2, b\lambda + a)$ lies on L:

$\frac{3\lambda}{1} = \frac{4\lambda+2}{2} = \frac{b\lambda+a-1}{1}$

$\lambda = 1, a + b - 1 = 3 \Rightarrow a + b = 4$ ...(1), $A = (4, 6, 4)$

Point $B(\mu + 1, 4\mu + 2, c\mu + a)$ lies on L:

$2\mu = 4\mu + 2 \Rightarrow \mu = -1, a - c - 1 = -1 \Rightarrow a = c$ ...(2), $B = (0, -2, 0)$

Since $PA = PB$ where $P = (1, 2, a)$:

$9 + 16 + (a - 4)^2 = 1 + 16 + a^2$

$\Rightarrow a = 3$, so $c = 3$, $b = 1$

$a+b+c=7$

14. (4) Find R by solving $L_1 = L_2$:

From $1+2\lambda = 4+5\mu$, $2+3\lambda = 1+2\mu$, $3+4\lambda = \mu$ we get $\lambda = -1$, $\mu = -1$, so $R = (-1,-1,-1)$

Point P on L_1: $P = (1+2\lambda_P, 2+3\lambda_P, 3+4\lambda_P)$ with $|\overrightarrow{PR}|^2 = 29$

$4(1+\lambda_P)^2 + 9(1+\lambda_P)^2 + 16(1+\lambda_P)^2 = 29 \rightarrow 29(1+\lambda_P)^2 = 29 \rightarrow \lambda_P = 0$ (first octant)

Thus $P = (1,2,3)$

Point Q on L_2: $Q = (4+5\mu_Q, 1+2\mu_Q, \mu_Q)$ with $|\overrightarrow{PQ}|^2 = \frac{47}{3}$

$(3+5\mu_Q)^2 + (-1+2\mu_Q)^2 + (\mu_Q-3)^2 = \frac{47}{3}$

$30\mu_Q^2 + 20\mu_Q + 19 = \frac{47}{3} \rightarrow 9\mu_Q^2 + 6\mu_Q + 1 = 0 \rightarrow \mu_Q = -\frac{1}{3}$

So $Q = (\frac{7}{3}, \frac{1}{3}, -\frac{1}{3})$

$|QR|^2 = \frac{100+16+4}{9} = \frac{120}{9} = \frac{40}{3}$

$27(QR)^2 = 360$

15. (6) Line L passes through P(1,1,1) perpendicular to two lines with directions $(4,1,1)$ and $(1,1,0)$.

Direction of L: $(4,1,1) \times (1,1,0) = (-1,1,3)$

Line L: $(x,y,z) = (1,1,1) + t(-1,1,3)$

Q is intersection with yz-plane (x=0): Setting $1-t=0$ gives $t=1$, so Q = (0, 2, 4)

Line through S(1,0,-1) parallel to L: $(x,y,z) = (1,0,-1) + s(-1,1,3)$

R is intersection with yz-plane: Setting $1-s=0$ gives $s=1$, so R = (0, 1, 2)

For parallelogram PQRS: $\vec{PQ} = (-1,1,3)$ and $\vec{PS} = (0,-1,-2)$

Area = $|\vec{PQ} \times \vec{PS}| = |(1,-2,1)| = \sqrt{1+4+1} = \sqrt{6}$

Square of area = 6